RAJALAKSHMI ENGINEERING COLLEGE

Sujath Ahamed S 250701757-2025-CSE S2

Started on	Friday, 14 August 2026, 3:48 PM
State	Finished
Completed on	Friday, 14 August 2026, 3:53 PM
Time taken	4 mins 37 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

void function(int n) {
    int c = 0;

    // Counter embedded in outer initialization and loop conditions
    for (int i = (c++, n / 2); (c++, i < n); i++) {
        for (int j = 1; (c++, j < n); j = 2 * j) {
            for (int k = 1; (c++, k < n); k = k * 2) {
                c++;
            }
        }
    }

    printf("%d\n", c);
}

int main() {
    int n;
    if (scanf("%d", &n) == 1) {
        function(n);
    }
    return 0;
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course